Measures - Mark scheme

Geometry · Edexcel IGCSE Mathematics A (4MA1) Paper 1 · Year 10

Question 1 *Practice question written for this worksheet (not from a past paper)*

The exchange rate is 1 pound = 4.60 dirhams.

(a) Change 35 pounds into dirhams.

(b) Change 920 dirhams into pounds.

(a) 35×4.60

(b) $920 \div 4.60$

Answer: (a) 161 dirhams (b) 200 pounds

Question 2 *Practice question written for this worksheet (not from a past paper)*

Change 2.4 m^2 into cm^2.

$1\ m^2 = 100 \times 100 = 10\,000\ cm^2$

$2.4 \times 10\,000$

Answer: 24 000 cm^2

Question 3 *Practice question written for this worksheet (not from a past paper)*

Change $3\,500\,000\ cm^3$ into m^3.

$1\ m^3 = 1\,000\,000\ cm^3$

$3\,500\,000 \div 1\,000\,000$

Answer: 3.5 m^3

Question 4 *Practice question written for this worksheet (not from a past paper)*

A film starts at 21:45 and lasts 2 hours 50 minutes.

At what time does it finish?

21:45 + 2 h = 23:45

23:45 + 50 min = 24:35 = 00:35

Answer: 00:35

Question 5 *Practice question written for this worksheet (not from a past paper)*

Change 15 m/s into km/h.

In 1 hour: $15 \times 3600 = 54\,000$ m

54 000 m = 54 km

Answer: 54 km/h

Question 6 *Practice question written for this worksheet (not from a past paper)*

A metal has density 8.4 g/cm^3. Work out the mass of 250 cm^3 of the metal.

Mass = density × volume

8.4×250

Answer: 2100 g

Question 7 *Practice question written for this worksheet (not from a past paper)*

A force of 600 N acts on an area of 0.5 m^2.

Work out the pressure.

Pressure = force ÷ area

600 ÷ 0.5

Answer: 1200 N/m²

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Question 8

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2026 · Question 12

Q	Working	Answer	Mark	Notes
	eg 5103 ÷ 7 **or** 5103 ÷ "420" (= 12.15) oe **or** 5103 ÷ "25 200" (= 0.2025) oe **or** "5 103 000" ÷ 7 (= 729 000) **or** "5 103 000" ÷ "420" (= 12 150) oe **or** "5 103 000" ÷ "25 200" (= 202.5) oe		2	M1 for use of speed = [distance] ÷ [time] where [distance] can be 5103 or 5103 × 1000 (= 5 103 000) and [time] can be 7 or 7 × 60 (= 420) or 7 × 60 × 60 (= 25 200)
	Correct answer scores full marks (unless from obvious incorrect working)	729		A1 cao
				Total 2 marks

Question 9

Edexcel IGCSE Mathematics A (4MA1) · Paper 2HR · June 2025 · Question 7

7		$10.5 = \frac{48.3}{v}$ **or** $10.5v = 48.3$ **or** $\left(v =\right) \frac{48.3}{10.5}$		2	M1 oe for substituting 10.5 and 48.3 correctly into a correct formula for density; may use any letter for the volume
		Correct answer scores full marks (unless from obvious incorrect working)	4.6		A1 allow $\frac{23}{5}$ **or** $4\frac{3}{5}$ oe
					Total 2 marks

Question 10

Practice question written for this worksheet (not from a past paper)

A box has a rectangular base 6 m by x m.

The force on the base is 1800 N and the pressure is 50 N/m².

Work out x.

Area = force ÷ pressure $= 1800 \div 50 = 36$

$6x = 36$

Answer: x = 6

Question 11

Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 9

9	e.g. 1.5 × 1.5 (= 2.25 oe)		3	M1	for calculating the area of the square, may be seen embedded within a calculation
	e.g. 34.8 × "2.25"			M1	for a complete method to find the force
		78.3		A1	oe
					Total 3 marks

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Question 12

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2024 · Question 23

Q	Working	Answer	Mark	Notes
	$1575 = (\text{area}) \times 21$ oe **or** (area =) 75 **or** $1575 = \pi \times r^2 \times 21$ oe **or** $r^2 = \frac{1575}{21\pi} \left(= 23.8(732...)\right)$ oe **or** $r = \sqrt{\frac{1575}{21\pi}} \left(= 4.88(602...)\right)$ oe		3	M1 for finding the area using Vol = cross sectional area × height **or** finding r or r^2 using vol = $\pi r^2 h$ NB r^2 and r can be rounded or truncated
	$\frac{84}{"75"}$ oe or $\frac{84}{\pi "4.88"^2}$ oe or $\frac{84}{\pi "23.8"}$ oe			M1 for $\frac{84}{\text{area of circle}}$
	Working not required, so correct answer scores full marks (unless from obvious incorrect working)	1.12		A1 accept 1.06 – 1.121
				Total 3 marks

Question 13

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2023 · Question 27

	$(V =) \frac{1950}{7.8} (= 250)$ **or** $7.8 = \frac{1950}{w \times 5 \times 4}$ or $7.8 = \frac{1950}{w \times 20}$		3	M1	for correct method to find volume using mass ÷ density or a correct equation with correct expression for volume (may be embedded in another calculation)
	eg $w = \frac{1950}{7.8 \times 5 \times 4}$ or $20w = \frac{1950}{7.8}$ or $20w =$ "250" or $4 \times 5 \times w =$ "250" **OR** eg $\frac{1950}{5 \times 4 \times 7.8}$ or $1950 \div (20 \times 7.8)$ or $1950 \div 156$ or "250" ÷ 20			M1	for a fully correct equation in w **or** a fully correct calculation to find the value of w (may be labelled eg x or L)
	Correct answer scores full marks (unless from obvious incorrect working)	12.5		A1	
					Total 3 marks

Question 14

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · June 2026 · Question 21

Question	Working	Answer	Mark	Notes
	eg $\frac{60}{24}(=2.5)$ oe **or** 150 minutes **or** 2 hours 30 minutes **or** $12 \times 1.25\ (=15)$ oe **or** $12 \times \frac{75}{60}(=15)$ oe		4	M1 for a method to work out the time taken to cycle 60 km in hours or minutes **or** for a method to work out the distance Henri runs Allow use of 1.15 or 1.4 for this mark
	eg $\frac{60}{24}(=2.5)$ oe **or** 150 minutes **or** 2 hours 30 minutes **AND** $12 \times 1.25\ (=15)$ oe **or** $12 \times \frac{75}{60}(=15)$ oe **OR** eg $\frac{60}{24}(=2.5)$ oe **or** 150 minutes **or** 2 hours 30 minutes **AND** 2:1 oe or 1:2 oe			M1 for a method to work out the time taken to cycle 60 km in hours or minutes **and** the distance Henri runs Allow use of 1.15 or 1.4 for this mark **OR** for a method to work out the time taken to cycle 60 km in hours and minutes **and** the ratio of times as 2:1
	$\frac{60+"15"}{"2.5"+1.25}$ oe **or** $\frac{75}{3.75}$ oe **or** $\frac{60+"15"}{"150"+75}(=0.333...)$ oe **or** $\frac{75}{225}$ oe **or** $\frac{2\times24+12}{3}$ oe			M1 for a method to find the average speed for the whole race. This may be in km/h or km/min Allow use of 1.15 or 1.4 for this mark
	Working required	20		A1 dep on M2 and correct use of time conversion
				Total 4 marks

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Question 15

Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · June 2025 · Question 22

Q	Working	Answer	Mark	Notes
(a)	$1262 \div 17.5$ oe		2	M1 $1262 \div$ their time their time may be an incorrect conversion to a decimal time eg 17.3 or from an attempt at converting to minutes eg 1050
		72		A1 accept 72.1 **or** 72.11...
(b)	$50x \div 1000 \ (= 0.05x)$ oe **or** $50x \times 60 \times 60 \ (= 180\,000x)$ oe **or** $50x \div \frac{1}{3600} \ (= 180\,000x)$ oe **or** $50x \div 1000 \times 60 \ (= 3x)$ **or** $\frac{3600}{1000}$ or $\frac{18}{5}$ or 3.6 **or** $\frac{1000}{3600}$ or $\frac{5}{18}$ or 0.277(77…)		3	M1 Condone omission of x for this marks
	$\frac{50x \times 60 \times 60}{1000}$ oe **or** $50x \times 3.6$ oe **or** $50x \div \frac{1000}{3600}$ oe **or** 180			M1 for a complete method including x **or** for an answer of 180
	Correct answer scores full marks (unless from obvious incorrect working)	$180x$		A1
				Total 5 marks

Question 16

Edexcel IGCSE Mathematics A (4MA1) · Paper 1F · November 2021 · Question 22

Q	Working	Answer	Mark	Notes
	$0.0027 = \frac{5.4}{(V)}$ oe		5	M1 for correctly using density $= \frac{\text{mass}}{\text{volume}}$
	$(V =) \frac{5.4}{0.0027} = 2000$			M1 for correctly rearranging for V
	$\pi \times 10^2 \times h = 2000$ oe			M1ft their 2000 for $\pi \times 10^2 \times h =$ their V
	$(h =) \frac{2000}{\pi \times 10^2}$ oe (= 6.3661…)			M1ft their 2000 dep on previous M1 for correctly rearranging for h
	Correct answer scores full marks (unless from obvious incorrect working)	6.4		A1 awrt 6.4
				Total 5 marks

CHALLENGE QUESTIONS

Question 17 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1FR · January 2022 · Question 3*

Q	Working	Answer	Mark	Notes
(a)		600	1	B1
(b)		4.5	1	B1
(c)	3 × 1000 (= 3000) **or** 225 ÷ 1000 (= 0.225)		4	M1
	"3000" ÷ 225 (= 13.3...) oe **or** 3 ÷ 0.225 (= 13.3...) oe			M1
	"3000" − ("13" × 225) **or** [3 − (13 × "0.225")] × 1000			M1 for a complete method
		75		A1
				Total 6 marks

Question 18 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · January 2022 · Question 7*

7	$220 \div 80$ (= 2.75 or $\frac{11}{4}$) oe			M1 for a method to find the time from B to C
	$72 \times \frac{50}{60}$ (= 60) oe			M1 for a method to find the distance from C to D Allow 0.83(333…) to 2 dp truncated or rounded
	$\frac{245+220+"60"}{2.5+"2.75"+\frac{50}{60}} \left(= \frac{525}{73\frac{1}{12}}\right)$ oe			M1 for a complete method to find the average speed for entire journey 0.83(333…) to 2 dp truncated or rounded 6.0(8333…) to 2 sf truncated or rounded
		86.3		A1 for 86.3 – 86.4
				Total 4 marks

Question 19 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · June 2022 · Question 7*

7	$1.4 = \frac{72}{(\text{area})}$ oe		4	M1
	(area =) $\frac{72}{1.4}$ $\left(= \frac{360}{7} = 51.4\ldots\right)$ oe			M1 (51.4 or better)
	"51.4…" × 18 **or** $r = \sqrt{\frac{"51.4..."}{\pi}}$ (= 4.046…) **and** $\pi \times "4.046"^2 \times 18$			M1 allow use of πr^2 to find the radius and then using $\pi r^2 h$ to find the volume
		926		A1 Allow 925 – 928
				Total 4 marks

Question 20 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1HR · November 2020 · Question 7*

7	Use of 2 hrs 42 mins = 2.7 hrs **or** 162 mins		4	B1
	e.g. 90 × 2.7 (= 243) **or** e.g. $\frac{90}{60} \times 162 (= 243)$ **or** e.g. $\frac{S}{90} = \frac{2.7}{3}$			M1 for use of $D = S \times T$ (accept use of their time e.g. 90 × 2.42) **or** for setting up an equation using proportion
	e.g. "243" ÷ 3 **or** $(S =)\ 90 \times \frac{2.7}{3}$			M1 (dep on M1) for their $D \div 3$ **or** for solving their equation
		81		A1
				Total 4 marks

Question 21 *Edexcel IGCSE Mathematics A (4MA1) · Paper 1H · January 2020 · Question 13*

13		$0.14 = \frac{56}{w^2}$ oe **or** 56 ÷ 0.14 (= 400)		4	M1 for using the given formula correctly
		$\sqrt{\frac{56}{0.14}}$ **or** $\sqrt{'400'}$ (=20)			M1 for a method to find w
		'20' × '20' × '20' oe			M1 (dep on M2) for a method to find the volume of the cube
			8000		A1
					Total 4 marks